



10/22/2022 – Sunday**

Student name: [redacted] Instructor: فايزة محمد Section: 11 → 2
النسب

[Helpful information: $R = 0.0821 \text{ atm.L/mol.K}$; mwt: $\text{H}_2\text{O} = 18 \text{ g/mol}$; $760 \text{ mmHg} = 1 \text{ atm}$]

* Choose the correct answer for the following questions and transfer it to the table below:

Question	1 ✓	2 ✓	3 ✓	4 ✓	5 ✓	6 ✓	7 A	8 ✓	9 ✓	10 B
Answer	D	A	C	C	A	A	B	A	D	D

Question	11 ✓	12 ✓	13 ✓	14 ✓	15 ✓	16 ✓	17 ✓	18 B	19 B	20 B
Answer	B	B	D	A	B	A	D	B	B	A

1- Which of the following statements is correct?

- A. When dealing with alcohols heat with direct flame.
- B. Return used chemicals to the stock bottles.
- C. Hold your face directly over the container when you smell a chemical substance.
- ☒ D. It's not allowed to withdraw samples from the stock solution bottles with your own pipet.

2-The following set of data were obtained by student during measurement density of solid material, Calculate Mass of the solid: -----:

- ☒ A. 1.33 kg
- B. 0.30 kg
- C. 3.00 g
- D. 13.3 g

$$d = \frac{m}{V}$$

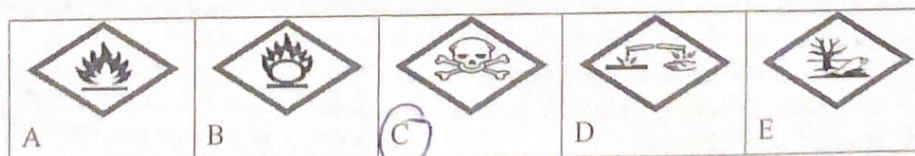
$$V = 16.5$$

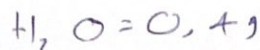
density of solid = 20 g/cm^3

Initial volume of water in cylinder: 18.5 ml

Final volume of water in cylinder: 85.0 ml

3-The pictogram that correctly indicates a Toxic chemical hazard is:





anhydrous = 0.7

alum = 1.12

4- Given the following set of data:

- Mass of empty crucible = 12.1g.	- Mass of crucible + alum = 13.2 0g.
- Mass of crucible + anhydrous salt = 12.8g	- Molar mass of anhydrous salt = 258.0g/mol.

Calculate water of hydration (X) in the alum.

A- 13

B- 14

C- 8

D- 5

5- If (X= 6) and you have the following data calculate mass of hydrates salt

A. 0.73 g

B. 1.75 g

C. 1.02 g

D. 0.68 g

Mass of empty crucible	5.21 g
Mass of crucible and anhydrous	6.23 g
Molar mass of anhydrous	150 g/mol
Molar mass of H ₂ O	18 g/mol

6- Calculate the temperature (in° C) of 1.4g of a vapor of a volatile Liquid (condensed vapor) (molar mass 90.0g/mol) which occupies a volume of 700.0 ml at a pressure of 720 mmHg:

A- 246 °C

B- 225°C

C- 519 C°

D- 158 °C

7- The boiling points in lab experiment are indicate using a

A. Capillary tube

B. Test tube

C. Thermometer

D. Bunsen burner

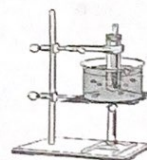
8- In which experiment dose this apparatus used:

A-Boiling point identification

C-Molecular weight of volatile liquid

~~B~~-Density identification

~~D~~-Water of crystallization



9- The vapor of an unknown volatile liquid occupies flask at 100.0°C and 1 atm, if the molar mass of volatile liquid equal 50.0 g/mol. what is the density of this volatile liquid?

A- 6.1 g/L

B- 1.2 g/L

C-1.9 g/L

D- 1.6 g/L

10- By heated to high temperature to drive off the water, this process called?

~~A~~ Boiling point

B. Dehydration

~~C~~ Hydrated

D. Crystallization

11- The boiling point of a liquid is dependent on pressure, which one of the following is correct

A. Yes, negative relationship with external pressure

B Yes, Positive relationship with external pressure

C. No, does not effected

D. No, negative relationship with vapor pressure

12- Which of the following statement is incorrect?

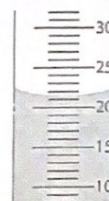
- ✓ A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ lower boiling point than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- ✗ B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ lower boiling point than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- ✓ C. CH_3OH higher boiling point than CH_3CH_3
- ✓ D. $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ higher boiling point than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

13- Which one of the following glass wear is more accurately?

- A. Beaker
- B. Graduated Cylinder
- C. Erlenmeyer Flask
- D. Burette

14- What is the volume inside the cylinder?

- A. 21.5 ml
- B. 22.5 ml
- C. 21.0 ml
- D. 22.0 ml



15- One of the following salt is insoluble in H_2O (polar solvent)?

- A. NaCl
- B. CaCO_3
- C. KI
- D. K_2SO_4

16- In which solvent is the ionic salt more soluble?

- A. Polar solvent
- B. Non-polar solvent
- C. Both polar and non-polar solvent
- D. None of the above

17- A method used to vaporize a liquid into a fixed volume vessel at a measured temperature and Atmospheric pressure to measured molecular weight is called?

- A. Synthesis method
- B. Condensation method
- C. Vaporization method
- D. Dumas method

18- During molecular weight of volatile liquid experiment. In the end of experiment, we weight.....and the volume of vapor take by using.....?(Hint's; cover: Al foil and rubber).

- ✗ A. mass of empty flask and cover - flask
- ✗ B. mass of flask, cover and vapor - flask
- C. mass of flask, cover and vapor - cylinder
- ✗ D. mass of empty flask and cover - cylinder

19- During the hydrated salt Experiment how value of X effected if we put sample in desiccator?

- A. Increase
- B. Decreases
- C. No effected

20- For most solid solutes, what happens to the solubility if the temperature decreases?

- A. Solubility increases
- B. Solubility decreases
- C. Solubility stays the same

Good luck